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# Heart Disease Prediction using ML

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# Tabular-dataset: UCI Heart Disease Data

Heart disease is one of the leading causes of death worldwide. Early detection is critical for effective treatment and prevention.

In this project, we aim to build machine learning models to predict the presence and severity of heart disease based on clinical and diagnostic features.

- Input: Patient medical attributes (age, chest pain type, ECG results, etc.)
- Output: Heart disease level (0–4)

Goal:

To analyze which features are most important and evaluate how well different models can predict heart disease.

[colab](#)

[Kaggle](#)

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# Dataset info:

|   | id | age | sex    | dataset   | cp              | trestbps | chol  | fbs   | restecg        | thalch | exang | oldpeak | slope       | ca  | thal              | num |
|---|----|-----|--------|-----------|-----------------|----------|-------|-------|----------------|--------|-------|---------|-------------|-----|-------------------|-----|
| 0 | 1  | 63  | Male   | Cleveland | typical angina  | 145.0    | 233.0 | True  | lv hypertrophy | 150.0  | False | 2.3     | downsloping | 0.0 | fixed defect      | 0   |
| 1 | 2  | 67  | Male   | Cleveland | asymptomatic    | 160.0    | 286.0 | False | lv hypertrophy | 108.0  | True  | 1.5     | flat        | 3.0 | normal            | 2   |
| 2 | 3  | 67  | Male   | Cleveland | asymptomatic    | 120.0    | 229.0 | False | lv hypertrophy | 129.0  | True  | 2.6     | flat        | 2.0 | reversable defect | 1   |
| 3 | 4  | 37  | Male   | Cleveland | non-anginal     | 130.0    | 250.0 | False | normal         | 187.0  | False | 3.5     | downsloping | 0.0 | normal            | 0   |
| 4 | 5  | 41  | Female | Cleveland | atypical angina | 130.0    | 204.0 | False | lv hypertrophy | 172.0  | False | 1.4     | upsloping   | 0.0 | normal            | 0   |

dataset- sample

- UCI Heart Disease Dataset
- 16 medical features
- Tabular data

# EDA - Numerical Features:

Age is approximately normally distributed

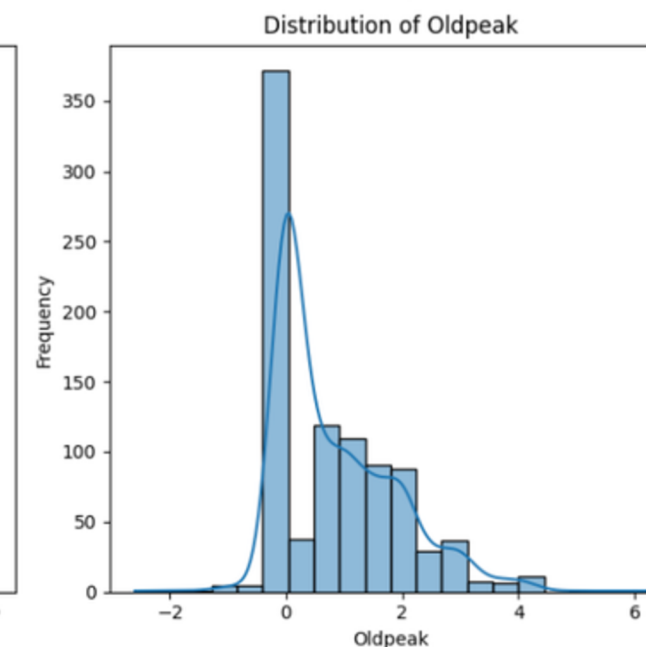
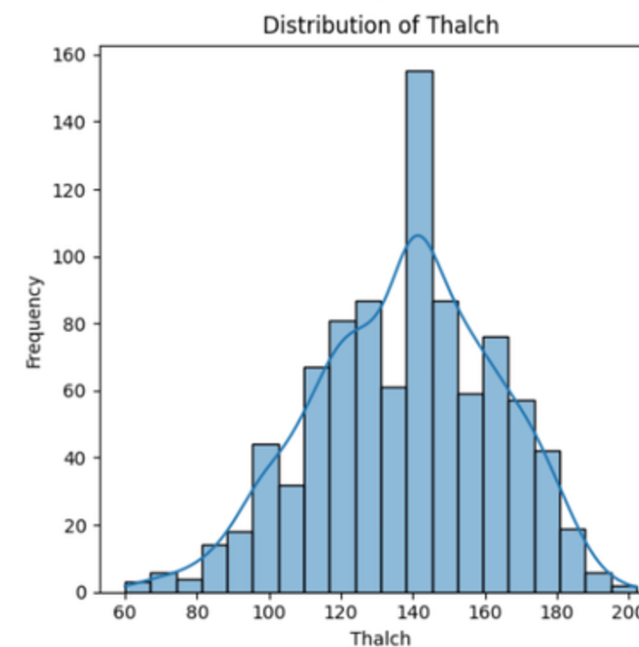
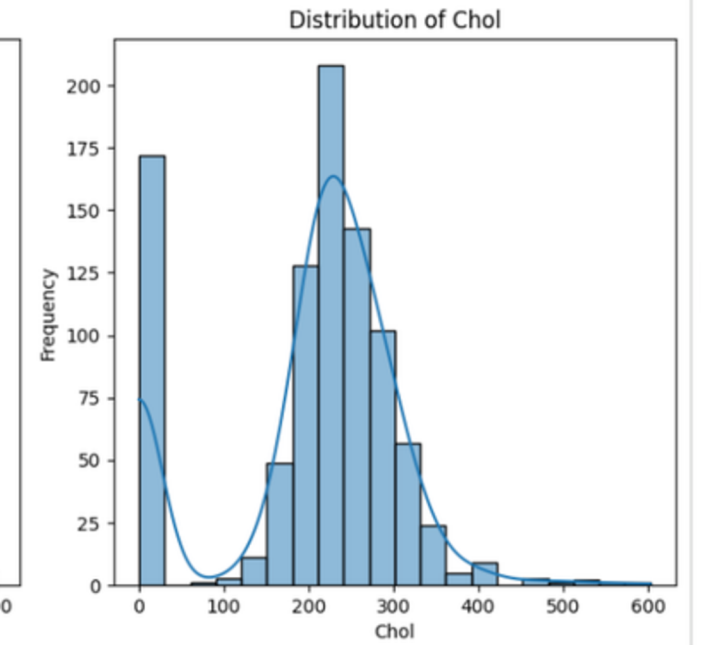
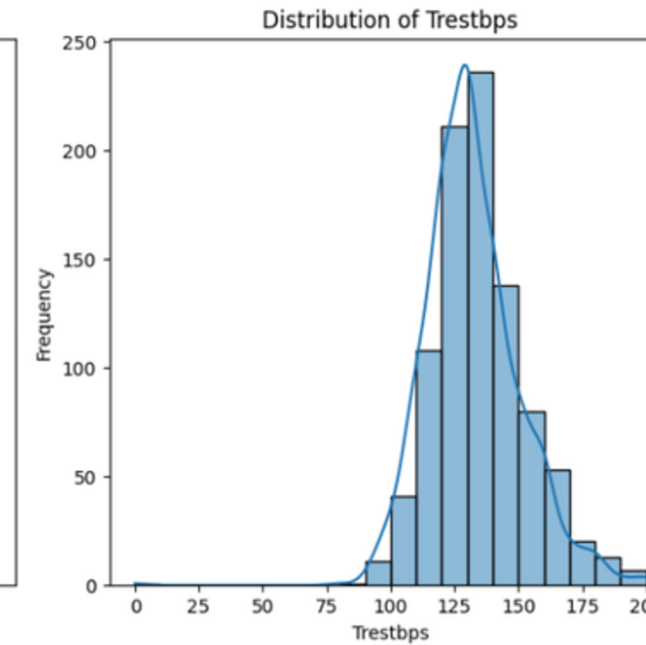
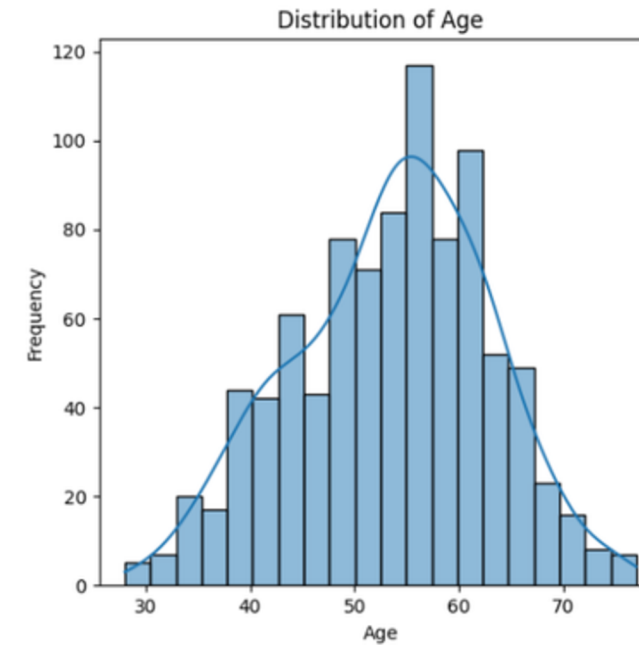
- Cholesterol (chol) shows strong right skew with outliers
- Oldpeak is highly skewed with many values near 0
- Resting blood pressure (trestbps) has moderate spread

Observations:

- Presence of outliers (especially in chol and oldpeak)
- Non-normal distributions in several features

Implication:

- Median imputation is more appropriate than mean
- Scaling is necessary for model stability



# EDA - Categorical Features:

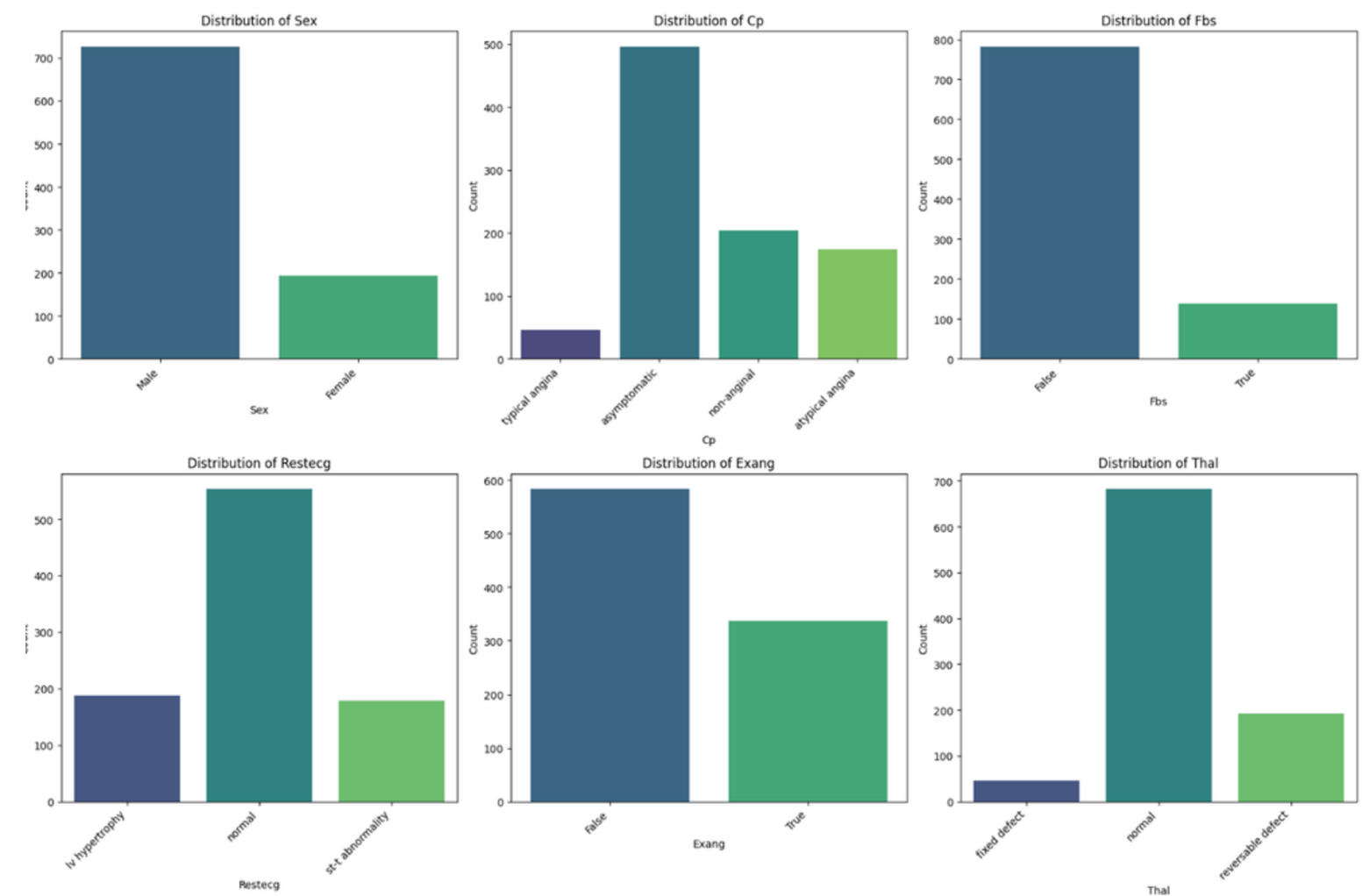
- Chest pain type (cp) shows strong correlation with heart disease
- Exercise-induced angina (exang) is highly associated with disease
- Thalassemia (thal) indicates significant risk patterns

## Key Insight:

- Asymptomatic chest pain → highest disease rate
- Reversible defects (thal) → strong indicator
- exang = True → higher probability of disease

## Weak features:

- fbs and restecg show weaker relationships



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# EDA Summary

- Heart disease prediction is influenced by multiple factors
- No single feature dominates → requires feature combination

Strong predictors:

- cp, exang, thal

Moderate predictors:

- age, thalach, oldpeak

Weak predictors:

- fbs, restecg

Key takeaway:

Models must capture interactions between features rather than relying on a single variable

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# Data Preprocessing

## Handling Missing Values

- Numerical features → filled with median
- Categorical features → filled with mode
- Dropped feature with excessive missing values (ca)

## Encoding

- Categorical variables → One-Hot Encoding
- Converted categories into binary features

## Feature Scaling

- Applied StandardScaler to numerical features
- Ensures consistent scale across variables

## Target Setup

- Multi-class classification (0–4)
  - Predicting different levels of heart disease severity
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# Models

- Logistic Regression → baseline
- SVM → decision boundary
- Random Forest → nonlinear model

| Model | Acc | F1 | AUC       |
|-------|-----|----|-----------|
| LR    | 57  | 54 | <b>82</b> |
| SVM   | 59  | 54 | 81        |
| RF    | 57  | 55 | 81        |

Although accuracy is moderate due to the multi-class nature of the problem, all models achieve ROC-AUC above 0.80, indicating strong ability to distinguish between different levels of heart disease.

In summary, the Random Forest model achieved the best overall performance among the tested models for predicting heart disease based on the F1-score and AUC metrics. While Logistic Regression offers good interpretability, the Random Forest provides superior predictive power by handling the inherent complexities and non-linearities within the clinical features. For practical application, the choice between these models would also depend on the specific requirements for interpretability versus predictive accuracy.

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